Q.29 Discuss Friction / Spin welding of plastics.

Q.30 Explain product life cycle and its stages.

Q.31 Discuss feasibility study and how it is important for any organization.

Q.32 Explain cost economics and its effect on plastic product design.

Q.33 Discuss tolerance and its types.

Q.34 Write short note on letters and alphabets used in plastic product design.

Q.35 Discuss various shapes and their importance.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain various preliminary design considerations for plastic product design.

Q.37 Discuss :

a) Discuss various processing limitations of polymer product deisgn

b) Hot gas and high frequency welding technique of plastic assembling

Q.38 Write short note on :

a) Explain various mechanical requirements of plastic product deisgn.

b) Explain various stages of product development

(20)

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6th Sem / Plastic, Chem Engg (SPL Polymer Tech)

Sub : Plastic Production Design

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Designs are periodically modified to ______.

a) Improve product performance

b) Strive for zero-based rejection and waste

c) Make products easier and faster to manufacture

d) All of the above

Q.2 PLC stands for _______.

a) Product life circle b) Product life cycle

c) Plastic life circle d) Product last cycle

Q.3 Method of adhesion, in which parent material is used as dope is known as _____

a) Solvent adhesion b) Solvent cementing

c) Solvent doping d) None of the above

Q.4 Alphabets and letters are generally used for _____

a) Identification of shift

b) For displaying company logos

c) For texturing effect

d) All of these

Q.5 _____ line is lying between two mould halves?

a) Parting line b) Centre line
c) Matching line d) Vertical line

Q.6 ______ must be avoided while designing mould component in blow moulding?

a) Radii b) Bend
c) Fillet d) Sharp corners

Q.7 The air of feasibility study is _____

a) To determine sources and profitability of organisation
b) To improve mould design
c) To give reaction in polymers
d) None of the above

Q.8 Name the solvent used for PMMA

a) MEK b) MCI
c) Benzene d) Ethanol

Q.9 The minimum distance between two consecutive holesis _____

a) d b) 2xd
c) 3xd d) 4xd

Q.10 Which the most difficult shape to prepare?

a) Artistic shapes b) Engineering shapes
c) Plain utility d) None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Name the different types of undercut.

Q.12 Draw any one insert for plastic molding.

Q.13 Name two permanent joining methods used in plastic assembly.

Q.14 Name two main types of moulded inserts.

Q.15 Give two advantages of Ribs.

Q.16 Name two types of holes used in plastic designs.

Q.17 Name various shapes used in plastic product designs.

Q.18 What is the drilling angle used in plastic products.

Q.19 Name two permanent joining methods used in plastic assembly.

Q.20 Line formed by meeting of two parallel flow fronts is known as ______.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Discuss various type of threads used in plastic product design.

Q.22 Explain gate side and its location.

Q.23 Discuss the causes and remedies for weld line.

Q.24 Explain Assembly methods.

Q.25 Suggest various plastic materials used for preparation of plastic gear.

Q.26 Discuss texturing and its importance.

Q.27 Explain various types of holes and their positioning with diagram.

Q.28 Discuss need and importance of uniform wall thickness of plastic product design.